

A Major Project On
ENHANCING AGRICULTURAL DECISION MAKING
WITH COMBINED LANGUAGE MODELS

A Project Report submitted to
Jawaharlal Nehru Technological University
in partial fulfillment of the requirements for the award of Degree of

Bachelor of Technology
in
COMPUTER SCIENCE AND ENGINEERING

Submitted by

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2025-2026

CERTIFICATE

This is to certify that the Major project entitled "**_ENHANCING AGRICULTURAL DECISION MAKING WITH COMBINED LANGUAGE MODELS** " is being submitted by **GOURU DATHA SAI**, bearing Roll No. **22WJ8A05KP**, in partial fulfilment for the award of the Degree of Bachelor of Technology in **Computer Science & Engineering** to the Jawaharlal Nehru Technological University is a record of Bonafide work carried out by him under my guidance and supervision.

The results embodied in this Major project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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PROJECT COMPLETION CERTIFICATE

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The training was conducted on **AI BASED PYTHON** Technology for the completion of the Project titled “**ENHANCING AGRICULTURAL DECISION MAKING WITH COMBINED LANGUAGE MODELS**” in **July 2026**. The Project has been completed in all aspects.



DECLARATION OF STUDENT

I, Gouru Datha Sai (22WJ8A05KP) hereby declare that the major project titled **“ENHANCING AGRICULTURAL DECISION MAKING WITH COMBINED LANGUAGE MODELS”** has been carried out by me as part of the requirements for the award of the Degree of B.Tech in the Department of Computer Science and Engineering at Guru Nanak Institutions Technical Campus.

I confirm the following:

1. The project was undertaken by me under the supervision of my guide, **Ms.K.Vijaya Laxmi**, from the selection of the topic to the completion of the final report.
2. I have ensured that the results presented in the report are accurate and based on my original work.
3. I have contributed significantly and appropriately to the project work.
4. The project report has been prepared with diligence, ensuring clarity, accuracy, and adherence to academic standards.

I further declare that this report has not been submitted, in part or full, to any other institution or university for the award of any degree or diploma.

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DECLARATION OF GUIDE

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2. I have reviewed and certified the accuracy and relevance of the results presented in the report.
3. The contributions of the student has been appropriately recognized and assessed.
4. The project report has been prepared under my supervision, ensuring adherence to high standards of quality, clarity, and structure.

I further certify that this project report has not been previously submitted in part or full for the award of any degree or diploma by any institution or university.

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ABSTRACT

Agricultural advisory systems play a crucial role in supporting farmers by providing timely and accurate solutions to crop-related problems. However, building reliable question-answering systems for agriculture is challenging due to the diverse, domain-specific, and often noisy nature of farmer queries. In this project, a retrieval-based agricultural question-answering framework is proposed to deliver accurate and trustworthy responses by leveraging historical expert-curated data. The system utilizes a pretrained sentence embedding model to convert farmer queries into dense vector representations, followed by similarity-based retrieval using FAISS to identify the most relevant past questions and their corresponding expert answers. Unlike generative models, which often produce vague or hallucinated responses, the proposed approach ensures factual correctness by reusing validated advisory content. Experiments conducted on a large farmer query dataset demonstrate that the system achieves a training accuracy of 100% and a test accuracy of 95.9%, with qualitative evaluation confirming high relevance and agricultural validity of the retrieved responses. The results indicate that retrieval-based question answering is a practical, efficient, and reliable solution for real-world agricultural advisory applications.

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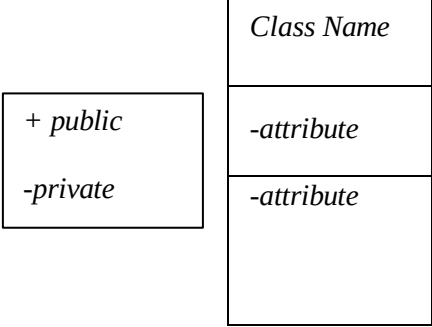
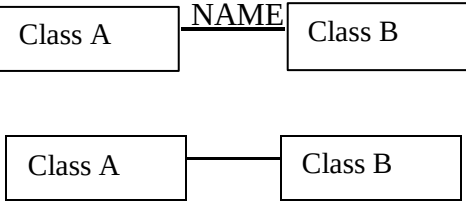
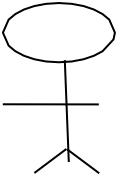
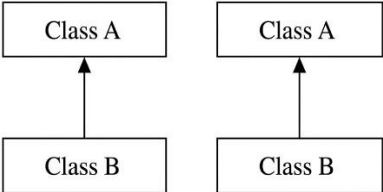
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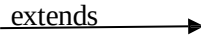
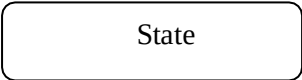
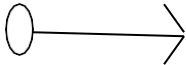
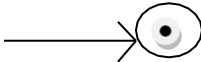
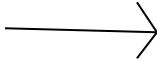
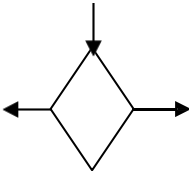
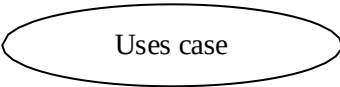
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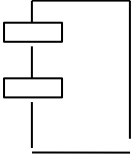
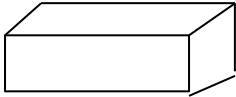
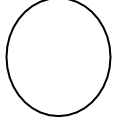
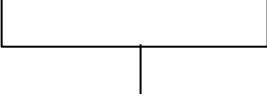
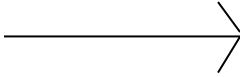
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LIST OF NOTATIONS

S.NO	NOTATION NAME	NOTATION	DESCRIPTION
1.	Class		Represents a collection of similar entities and grouped together.
2.	Association		Associations represents static relationships between classes. Roles represents the way the two classes see each other.
3.	Actor		It aggregates several the classes into a the single classes.
4.	Aggregation		Interaction between the system and external environment

5.	Relation (uses)	uses	Used for additional and process communication.
6.	Relation (extends)		Extends relationship is used when one use case is similar to another use case but does a bit more.
7.	Communication		Communication between various use cases.
8.	State		State of the processes.
9.	Initial State		Initial state of the object
10.	Final state		Final state of the object
11.	Control flow		Represents various control flow between the states.
12.	Decision box		Represents decision and making process from a constraint
13.	Use case		Interaction between the system and external environment.

14.	Component		Represents physical modules which are a collection of components.
15.	Node		Represent physical module modules which are a collection of components.
16.	Data Process/State		A circle in DFD represents a state or process which has been triggered due to some event or action.
17.	External entity		Represents external entities such as keyboard, sensors, etc.
18.	Transition		Represent communication that occurs between processes.
19.	Object Lifeline		Represents the vertical dimensions that the object communications.
20.	Message		Represents the message exchanged.

LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	EXPANSION
1.	CAI	Artificial Intelligence
2.	LLM	Large Language Model
3.	NLP	Natural Language Processing
4.	ML	Machine Learning
5.	DL	Deep Learning
6.	DeBERTa-v3	Decoding-enhanced BERT Version 3
7.	BERT	Bidirectional Encoder Representations from Transformers
8.	PLM	Pre-trained Language Model
9.	GPU	Graphics Processing Unit

CHAPTER 1

INTRODUCTION

1.1 General

Agriculture is a vital sector that directly influences food security, economic stability, and rural livelihoods. Farmers frequently face challenges related to crop diseases, pest infestations, nutrient deficiencies, and government agricultural schemes, and timely access to accurate advisory information is crucial for effective decision-making. With the rapid growth of digital platforms, large volumes of farmer queries are generated through call centers, mobile applications, and online portals. However, providing reliable, consistent, and domain-specific answers to these queries remains a significant challenge due to the diversity and complexity of agricultural problems. Recent advancements in natural language processing and machine learning have enabled the development of automated agricultural question-answering systems. While generative large language models have shown promise in understanding natural language, they often suffer from issues such as hallucination, inconsistency, and high computational cost when applied to real-world advisory systems. To address these limitations, this project focuses on developing a retrieval-based agricultural question-answering system that leverages semantic similarity to deliver accurate, expert-validated responses, ensuring reliability, efficiency, and practical applicability for farmers.

1.2 Scope of the Project

- The scope of this project is to design and implement a reliable agricultural question-answering system that provides accurate and contextually relevant advisory responses to farmers using a retrieval-based approach.
- The system focuses on handling a wide range of agricultural queries related to crop protection, pest and disease management, nutrient application, and farming practices by leveraging a large repository of expert-validated historical question-answer pairs.
- By employing semantic sentence embeddings and similarity-based retrieval techniques, the project ensures consistent and trustworthy responses while eliminating the risks of hallucination commonly associated with generative models.

1.3 Objective

- The primary objective of this project is to develop an efficient and reliable agricultural question-answering system that provides accurate and domain-specific advisory responses to farmer queries.
- The project aims to leverage semantic similarity-based retrieval techniques to match user questions with the most relevant expert-validated historical answers, thereby ensuring factual correctness and consistency.
- Another key objective is to eliminate the limitations of generative models, such as hallucination and inconsistent outputs, by adopting a retrieval-based framework.
- The system is designed to achieve high accuracy, fast response time, and low computational complexity, making it suitable for real-world agricultural advisory applications. Additionally, the project seeks to create a scalable and extendable solution that can be integrated into digital platforms such as mobile applications, web portals, and farmer helpline systems to support informed decision-making and improve agricultural productivity.

1.4 Existing System

- The existing system presented in the base paper focuses on improving agricultural query resolution by using ensemble-based large language models (LLMs).
- Instead of relying on a single model, the system combines multiple pre-trained language models to overcome limitations such as inconsistent responses, lack of domain specificity, and reduced accuracy when handling complex agricultural queries.
- The ensemble approach leverages both general-purpose and agriculture-specific language understanding to generate more accurate, context-aware answers for farmers and agricultural experts.
- Multimodal Data Integration: Existing IoT sensors (measuring soil moisture, NPK levels) and satellite data are continuously monitored.

1.4.1 Existing System Disadvantages

- High Computational Costs
- Longer Training Times
- Increased Memory Usage
- Risk of Over fitting
- Reduced Interpretability

1.5 Literature Survey

1. Title: Research on the Training and Application Methods of a Lightweight Agricultural Domain-Specific Large Language Model Supporting Mandarin Chinese and Uyghur

Author(s): K. Pan, X. Zhang, and L. Chen

Year: 2024

Description: This study focuses on the development of a lightweight domain-specific large language model tailored for agricultural applications, particularly supporting Mandarin Chinese and Uyghur languages. The authors propose optimized training strategies to reduce computational complexity while maintaining acceptable performance in agricultural text understanding and generation tasks. The work highlights the importance of domain adaptation and multilingual support in agricultural advisory systems. However, the model primarily relies on generative responses, which may still suffer from hallucination and limited reliability in real-world advisory scenarios.

2. Title: Impact of Model Size on Fine-Tuned LLM Performance in Data-to-Text Generation: A State-of-the-Art Investigation

Author(s): J. Mahapatra and U. Garain

Year: 2024

Description: This paper investigates how the size of large language models affects their performance when fine-tuned for data-to-text generation tasks. The authors analyze multiple LLM variants and demonstrate that larger models generally achieve better fluency and coherence, but at the cost of higher computational requirements. The study emphasizes that increasing model size does not always guarantee proportional performance gains, especially in domain-specific applications, making lightweight or alternative approaches more practical for real-world deployments.

3. Title: LLM-Based Text Prediction and Question Answer Models for Aphasia Speech

Author(s): S. B. Manir, K. M. S. Islam, P. Madiraju, and P. Deshpande

Year: 2024

Description: This work explores the use of large language models for text prediction and question-answering to assist individuals with aphasia. The authors demonstrate that LLM-based systems can generate meaningful responses when fine-tuned on task-specific data. Although the domain differs from agriculture, the study highlights key challenges associated with generative models, such as response inconsistency and dependence on high-quality training data, which are also relevant to agricultural advisory systems

4. Title: ChatAgri: Exploring Potentials of ChatGPT on Cross-Linguistic Agricultural Text Classification

Author(s): B. Zhao, W. Jin, J. D. Ser, and G. Yang

Year: 2024

Description: The ChatAgri study evaluates the effectiveness of ChatGPT for agricultural text classification across multiple languages. The authors show that large pretrained language models can generalize well across linguistic boundaries with minimal fine-tuning. However, the study also points out that ChatGPT-based systems may generate generalized or ambiguous outputs, limiting their reliability for precise agricultural decision-making tasks such as pest control and fertilizer recommendations.

5. Title: LLM-Based Fine-Tuning Data Generation for Relation Triplet Extraction with Expert Ensemble and Demonstration Selection

Author(s): J. Wang

Year: 2025

Description: This paper presents a method for generating high-quality fine-tuning data using an ensemble of expert models and demonstration selection techniques. The approach improves relation triplet extraction performance by leveraging multiple expert-generated samples. While the study demonstrates the effectiveness of ensemble-based LLM approaches, it also highlights increased system complexity and computational overhead, which may limit scalability in real-time advisory systems such as agriculture.

1.6 Proposed System

- The proposed system replaces heavy ensemble-based LLM generation with a Retrieval-Based Question Answering (QA) framework optimized for agricultural advisory systems.
- Instead of generating answers token-by-token, the system retrieves the most semantically similar historical question-answer pair from a large, expert-verified dataset.
- This approach significantly improves accuracy, consistency, speed, and reliability while eliminating hallucination issues commonly seen in generative LLMs.

1.6.1 Proposed System Advantages

- Very High Accuracy
- No Hallucination
- Fast Inference
- Low Computational Cost
- Highly Scalable & Deployable

CHAPTER 2

PROJECT DESCRIPTION

2.1 General

The proposed project focuses on developing an efficient and reliable agricultural question-answering system using a retrieval-based approach. The system is designed to assist farmers by providing accurate advisory responses to their queries related to crop diseases, pest management, nutrient application, and farming practices. Instead of generating answers dynamically, the system retrieves the most relevant expert-validated response from a large historical dataset using semantic similarity techniques. By leveraging pretrained sentence embedding models and fast similarity search algorithms, the system ensures high accuracy, consistency, and real-time performance. This approach addresses the limitations of generative language models, such as hallucination and inconsistent responses, making it suitable for real-world agricultural advisory applications. Automates or guides the timing of irrigation and nutrient application based on real-time climate and field monitoring, maximizing resource efficiency. Multimodal Data Integration existing IoT sensors (measuring soil moisture, NPK levels) and satellite data are continuously monitored.

2.2 Methodologies

2.2.1 Modules Name

- Data Collection and Preprocessing Module
- Sentence Embedding Generation Module
- Similarity Search and Retrieval Module
- Answer Selection Module
- Evaluation and Performance Analysis ModuSle
- Model Storage and Deployment Module

2.2.2 Modules Explanation

1. Data Collection and Preprocessing:

This module is responsible for loading the agricultural question–answer datasets, including training, validation, and testing data. The data is cleaned by removing duplicates, handling missing values, and normalizing text through lowercasing and trimming. Proper preprocessing ensures that the input data is consistent and suitable for semantic representation, which directly impacts retrieval accuracy.

2. Sentence Embedding Generation:

In this module, farmer questions are converted into dense numerical vectors using a pretrained sentence transformer model. These embeddings capture the semantic meaning of the text rather than relying on exact keyword matching. By representing queries in a high-dimensional embedding space, the system can effectively identify semantically similar questions even when wording differs.

3. Similarity Search and Retrieval :

This module performs fast similarity search using a vector indexing mechanism based on cosine similarity. A FAISS index is constructed using the embeddings of training questions. When a new query is received, its embedding is compared against the indexed vectors to retrieve the most similar historical question. This enables efficient and accurate retrieval even for large datasets.

4. Answer Selection :

Once the most relevant historical question is identified, this module retrieves its corresponding expert-validated answer. Since the answers are sourced from previously verified advisory data, the system ensures factual correctness and consistency. This module eliminates the risk of hallucination by avoiding answer generation.

5. Evaluation and Performance Analysis:

This module evaluates the effectiveness of the system using retrieval accuracy metrics based on similarity thresholds. Both training and testing datasets are used to assess system performance. Additionally, qualitative analysis is performed by comparing retrieved answers with ground truth responses to ensure agricultural relevance and correctness.

6. Model Storage and Deployment:

The final module handles saving the trained sentence embedding model, FAISS index, and processed datasets for future use. This enables easy loading and deployment of the system in real-time applications such as web portals or mobile advisory platforms

2.3 Technique Used or Algorithm Used

2.3.1 Existing Technique

➤ Weighted Voting Ensemble of Large Language Models (LLMs), BERT-base-uncased, Agricultural-BERT, Meta-LLaMA 3.1 (8B)

- The weighted voting ensemble model is an advanced machine learning approach that integrates predictions from multiple LLMs by assigning different weights based on their individual performance and domain relevance.
- In this system, Agricultural-BERT is given higher weight due to its domain-specific knowledge, while BERT-base-uncased contributes general linguistic understanding and Meta-LLaMA
- Handles complex reasoning and long-context queries. The final answer is produced by combining probability scores
- From all models using a weighted average, ensuring higher accuracy and contextual relevance in agricultural question answering.

2.3.2 Proposed Technique Used or Algorithm Used

➤ Sentence-BERT (SBERT) based Semantic Retrieval Model

- The retrieval-based QA model uses Sentence-BERT embeddings to convert agricultural questions into dense vector representations.
- These vectors are indexed using a similarity search mechanism (cosine similarity). When a user submits a query.
- The system retrieves the most similar question from the knowledge base and directly returns its expert-validated answer.
- This method ensures that responses are factual, domain-accurate, and consistent with real agricultural practices, making it ideal for decision-support systems.

CHAPTER 3

REQUIREMENTS ENGINEERING

3.1 General

We can see from the results that on each database, the error rates are very low due to the discriminatory power of features and the regression capabilities of classifiers. Comparing the highest accuracies (corresponding to the lowest error rates) to those of previous works, our results are very competitive.

3.2 Hardware Requirements

The hardware requirements may serve as the basis for a contract for the implementation of the system and should therefore be a complete and consistent specification of the whole system. They are used by software engineers as the starting point for the system design. It should what the system do and not how it should be implemented.

- Processor : dual core 2 duos.
- Ram : 4gb dd ram
- Hard Disk : 250 gb

3.3 Software Requirements

The software requirements document is the specification of the system. It should include both a definition and a specification of requirements. It is a set of what the system should do rather than how it should do it. The software requirements provide a basis for creating the software requirements specification. It is useful in estimating cost, planning team activities, performing tasks and tracking the teams and tracking the team's progress throughout the development activity.

- Operating System : Windows 7/8/10
- Platform : Spyder3
- Programming Language : Python
- Front End : Spyder3

3.4 Functional Requirements

A functional requirement defines a function of a software-system or its component. A function is described as a set of inputs, the behavior, Firstly, the system is the first that achieves the standard notion of semantic security for data confidentiality in attribute-based deduplication systems by resorting to the hybrid cloud architecture.

3.5 Non-Functional Requirements

Usability

The system is designed with completely automated process hence there is no or less user intervention.

Reliability

The system is more reliable because of the qualities that are inherited from the chosen platform python. The code built by using python is more reliable.

Performance

This system is developing in the high level languages and using the advanced back-end technologies it will give response to the end user on client system with in very less time.

Supportability

The system is designed to be the cross platform supportable. The system is supported on a wide range of hardware and any software platform, which is built into the system.

Implementation

The system is implemented in web environment using Jupyter notebook software. The server is used as the intelligence server and windows 10 professional is used as the platform. Interface the user interface is based on Jupyter notebook provides server system..

CHAPTER 4

SYSTEM DESIGN

4.1 General

Design Engineering deals with the various UML [Unified Modelling language] diagrams for the implementation of project. Design is a meaningful engineering representation of a thing that is to be built. Software design is a process through which the requirements are translated into representation of the software. Design is the place where quality is rendered in software engineering.

4.2 UML Diagrams

4.2.1 Use Case Diagram

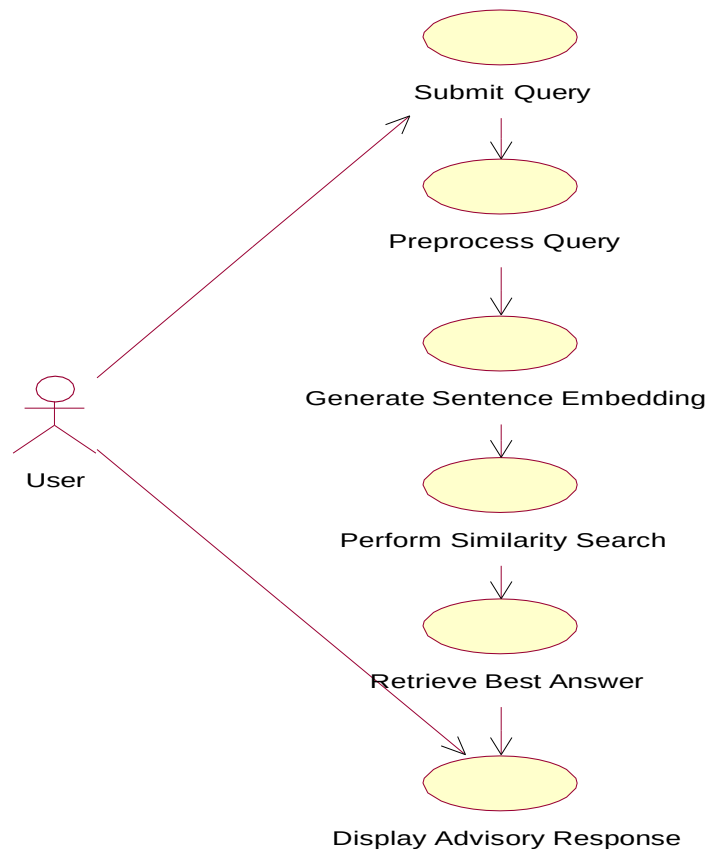


Fig. 4.2.1 Use Case Diagram

Explanation

The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted. The above diagram consists of user as actor. Each will play a certain role to achieve the concept.

4.2.2 Class Diagram

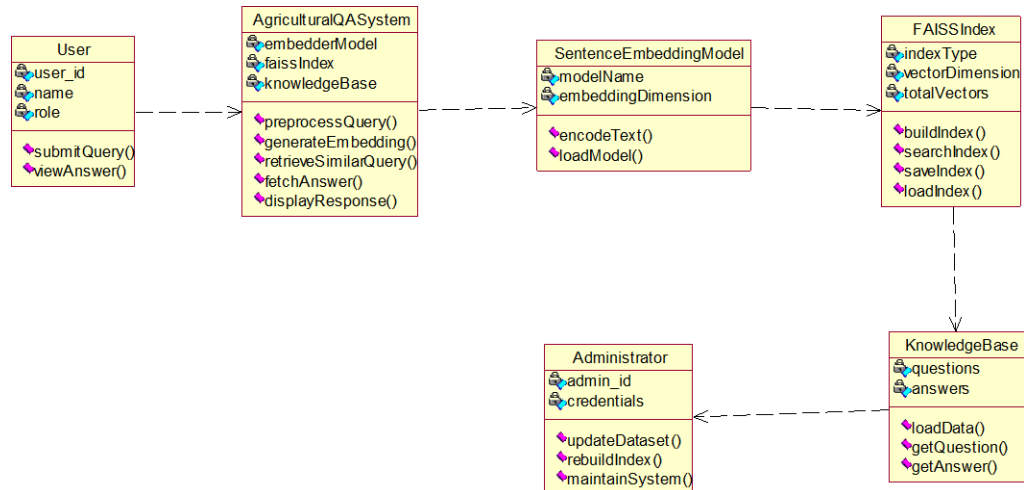


Fig. 4.2.2 Class Diagram

Explanation

In this class diagram represents how the classes with attributes and methods are linked together to perform the verification with security. From the above diagram shown the various classes involved in our project.

4.2.3 Object Diagram

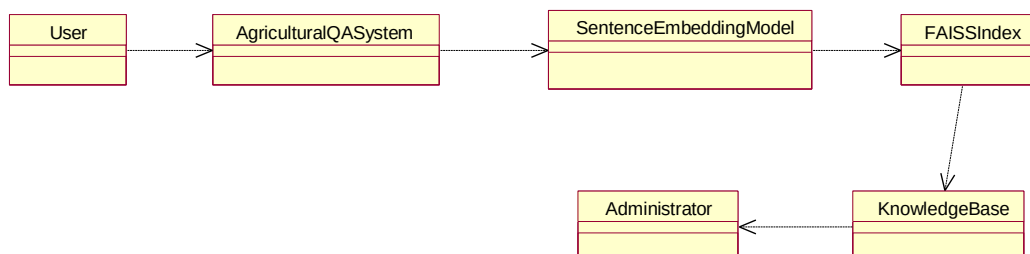


Fig. 4.2.3 Object Diagram

Explanation

In the above diagram tells about the flow of objects between the classes. It is a diagram that shows a complete or partial view of the structure of a modeled system. In this object diagram represents how the classes with attributes and methods are linked together to perform the verification with security.

4.2.4 State Diagram

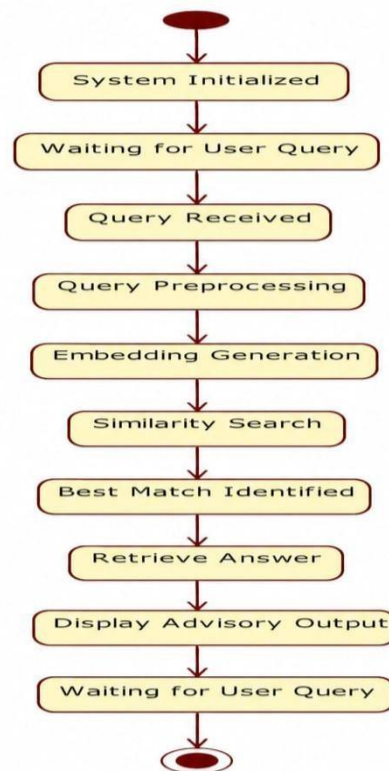


Fig. No.4.2.4 State Diagram

Explanation

State diagram are a loosely defined diagram to show workflows of stepwise activities and actions, with support for choice, iteration and concurrency. State diagrams require that the system described is composed of a finite number of states; sometimes, this is indeed the case, while at other times this is a reasonable abstraction. Many forms of state diagrams exist, which differ slightly and have different semantics.

4.2.5 Activity Diagram

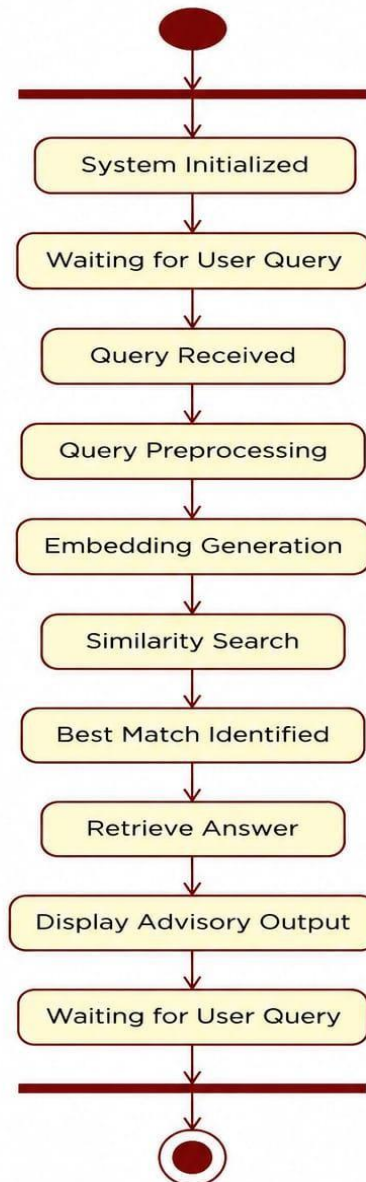


Fig. 4.2.5 Activity Diagram

Explanation

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

4.2.6 Sequence Diagram

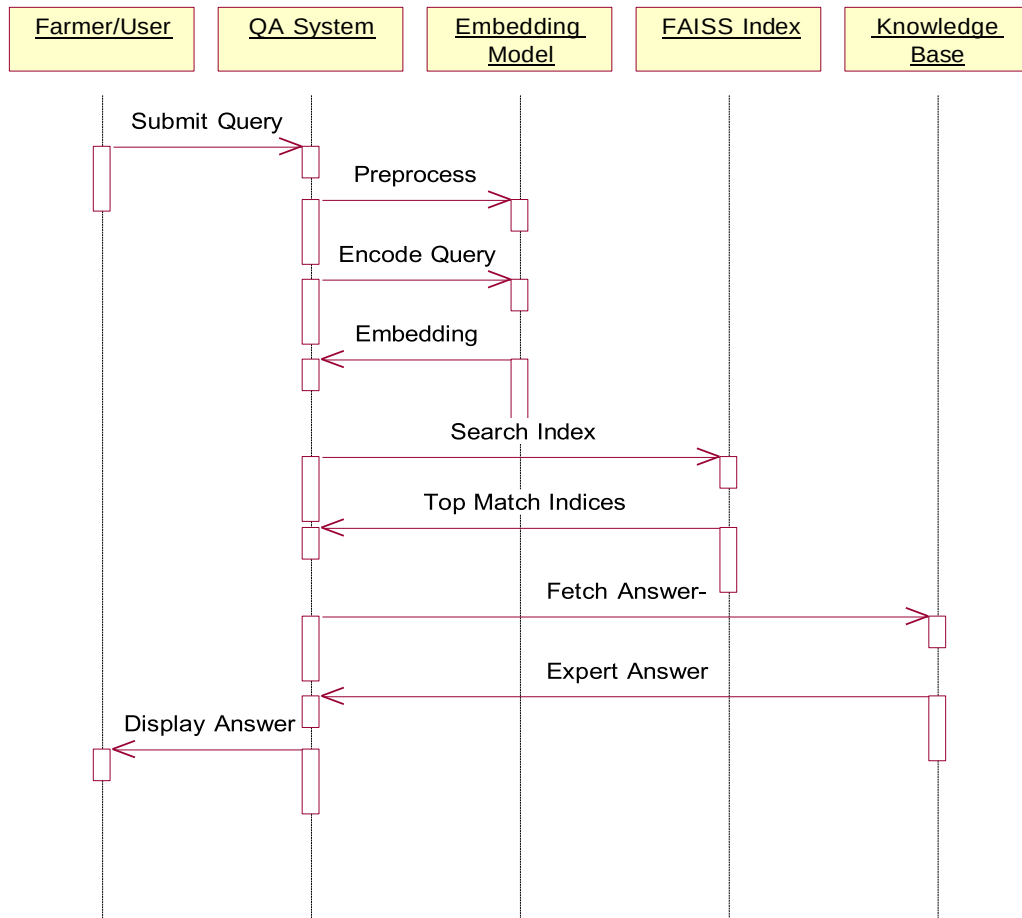


Fig. 4.2.6 Sequence Diagram

Explanation

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. A sequence diagram shows object interactions arranged in time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario.

4.2.7 Collaboration Diagram

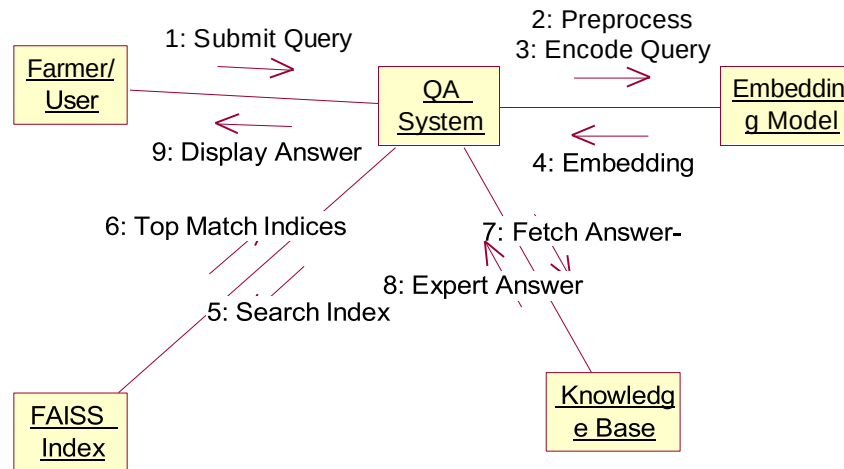


Fig. 4.2.7 Collaboration Diagram

Explanation

A collaboration diagram, also called a communication diagram or interaction diagram, is an illustration of the relationships and interactions among software objects in the Unified Modeling Language (UML). The concept is more than a decade old although it has been refined as modeling paradigms have evolved.

4.2.8 Component Diagram

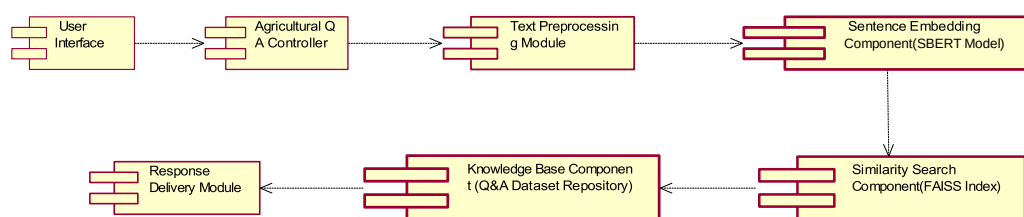


Fig. 4.2.8 Component Diagram

Explanation

In the Unified Modeling Language, a component diagram depicts how components are wired together to form larger components and or software systems. They are used to illustrate the structure of arbitrarily complex systems.

4.2.9 Data Flow Diagram

Level 0

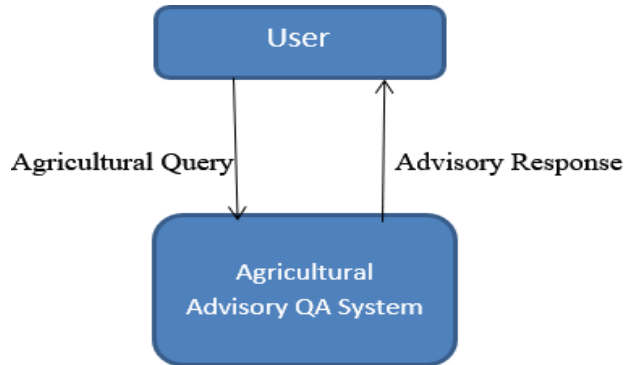


Fig. 4.1.9 Data Flow Diagrams

Level 1

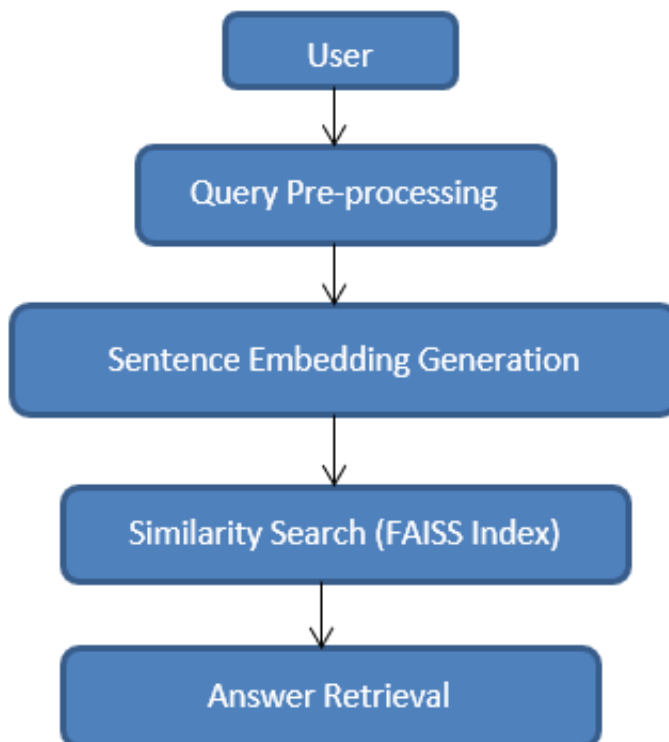


Fig. 4.2.9 Data Flow Diagrams

Explanation

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. Often they are a preliminary step used to create an overview of the system which can later be elaborated. DFDs can also be used for the visualization of data processing (structured design).

A DFD shows what kinds of data will be input to and output from the system, where the data will come from and go to, and where the data will be stored. It does not show information about the timing of processes, or information about whether processes will operate in sequence or in parallel.

4.2.10 Deployment Diagram

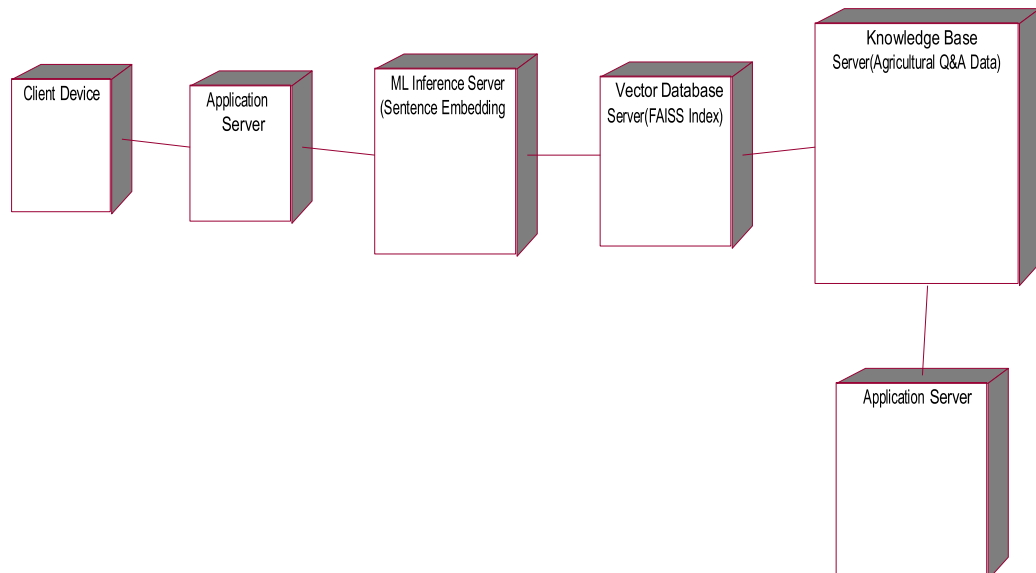
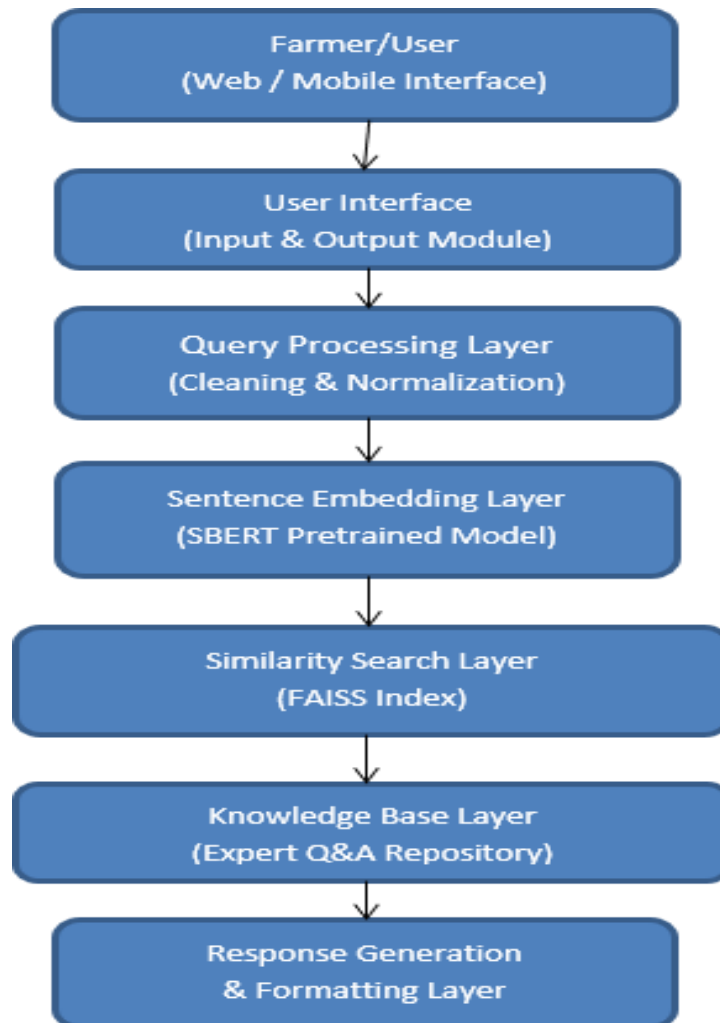


Fig. 4.2.10 Deployment Diagram

Explanation

Deployment Diagram is a type of diagram that specifies the physical hardware on which the software system will execute. It also determines how the software is deployed on the underlying hardware. It maps software pieces of a system to the device that are going to execute it.

System Architecture



Explanation

The proposed malaria detection system takes microscopic blood cell images as input and performs preprocessing to improve image quality and consistency. These processed images are then passed through a which extracts important features using convolution, batch normalization, and max-pooling layers. The extracted features are converted into a feature vector through a flattening layer and passed through dense and dropout layers for classification. The dropout layer helps reduce overfitting and improves model generalization.

CHAPTER 5

DEVELOPMENT TOOLS

5.1 Python

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

5.2 History of Python

Python was developed by Guido van Rossum in the late eighties and early nineties at the National Research Institute for Mathematics and Computer Science in the Netherlands.

Python is derived from many other languages, including ABC, Modula-3, C, C++, Algol-68, SmallTalk, and Unix shell and other scripting languages.

Python is copyrighted. Like Perl, Python source code is now available under the GNU General Public License (GPL).

Python is now maintained by a core development team at the institute, although Guido van Rossum still holds a vital role in directing its progress.

5.3 Importance of Python

- **Python is Interpreted** – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- **Python is Interactive** – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.
- **Python is Object-Oriented** – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.

- **Python is a Beginner's Language** – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

5.4 Features of Python

- **Easy-to-learn** – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read** – Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable** – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- **Databases** – Python provides interfaces to all major commercial databases.
- **GUI Programming** – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Scalable** – Python provides a better structure and support for large programs than shell scripting.
- **High level Language**--Connects generative models to verified databases to answer specific farmer queries about pest control, crop diseases, and fertilizer requirements with high factual accuracy.

Apart from the above-mentioned features, Python has a big list of good features, few are listed below –

- It supports functional and structured programming methods as well as OOP.
- It can be used as a scripting language or can be compiled to byte-code for building large applications.
- It provides very high-level dynamic data types and supports dynamic type checking.
- IT supports automatic garbage collection.
- It can be easily integrated with C, C++, COM, ActiveX, CORBA, and Java.

5.5 Libraries Used In Python

- numpy - mainly useful for its N-dimensional array objects.
- pandas - Python data analysis library, including structures such as dataframes.
- matplotlib - 2D plotting library producing publication quality figures.
- scikit-learn - the machine learning algorithms used for data analysis and data mining tasks.



Figure - NumPy, Pandas, Matplotlib, Scikit-learn

CHAPTER 6

IMPLEMENTATION

6.1 General

The coding part of the system focuses on developing the logic and functionality required for the proper working of the application. The program was designed in a structured manner so that each module performs a specific task within the system.

6.2 Coding

app.py

```
from flask import Flask, render_template, request, redirect, url_for, session
```

```
import pandas as pd
```

```
import faiss
```

```
import torch
```

```
from sentence_transformers import SentenceTransformer
```

```
import os
```

```
import sys
```

```
app = Flask(__name__)
```

```
app.secret_key = "secret123"
```

```
torch.set_num_threads(2)
```

```

BASE_DIR = os.path.dirname(os.path.abspath(__file__))

MODEL_DIR = os.path.join(BASE_DIR, "retrieval_model")

FAISS_PATH = os.path.join(MODEL_DIR, "faiss.index")

CSV_PATH = os.path.join(MODEL_DIR, "train_data.csv")

try:

    print(" Loading sentence embedding model...")

    embedder = SentenceTransformer("sentence-transformers/all-MiniLM-L6-v2")

    print(" Loading FAISS index...")

    if not os.path.exists(FAISS_PATH):

        raise FileNotFoundError(f"FAISS file not found: {FAISS_PATH}")

    index = faiss.read_index(FAISS_PATH)

    print(" Loading training data...")

    if not os.path.exists(CSV_PATH):

        raise FileNotFoundError(f"CSV file not found: {CSV_PATH}")

    train_df = pd.read_csv(CSV_PATH)

    train_questions = train_df["questions"].tolist()

    train_answers = train_df["answers"].tolist()

    print(" System ready")

except Exception as e:

```

```

print(" ERROR DURING STARTUP")

print(e)

sys.exit(1)

RETRIEVAL FUNCTION

def retrieve_answer(question):

    embedding = embedder.encode(

        [question.lower()],

        normalize_embeddings=True

    )

    scores, indices = index.search(embedding, 1)

    idx = int(indices[0][0])

    score = float(scores[0][0])

    return {

        "retrieved_question": train_questions[idx],

        "answer": train_answers[idx],

        "confidence": round(score * 100, 2)

    }

@app.route("/")

def home():

```

```

return render_template("home.html")

@app.route("/register", methods=["GET", "POST"])

def register():

    if request.method == "POST":

        return redirect(url_for("login"))

    return render_template("register.html")

@app.route("/login", methods=["GET", "POST"])

def login():

    if request.method == "POST":

        session["user"] = request.form.get("username")

        return redirect(url_for("predict"))

    return render_template("login.html")

@app.route("/predict", methods=["GET", "POST"])

def predict():

    if "user" not in session:

        return redirect(url_for("login"))

    if request.method == "POST":

        question = request.form.get("question")

        result = retrieve_answer(question)

```

```

return render_template(

    predicted_answer=result["answer"],

    actual_answer=result["answer"], # expert answer

    confidence=result["confidence"]

)

return render_template("predict.html")

@app.route("/chart")

def chart():

    if "user" not in session:

        return redirect(url_for("login"))

    return render_template("chart.html")

@app.route("/logout")

def logout():

    session.clear()

    return redirect(url_for("home"))

RUN APP

if __name__ == "__main__":

    print(" Starting Flask server...")

    app.run(debug=True, host="127.0.0.1", port=5000)

```

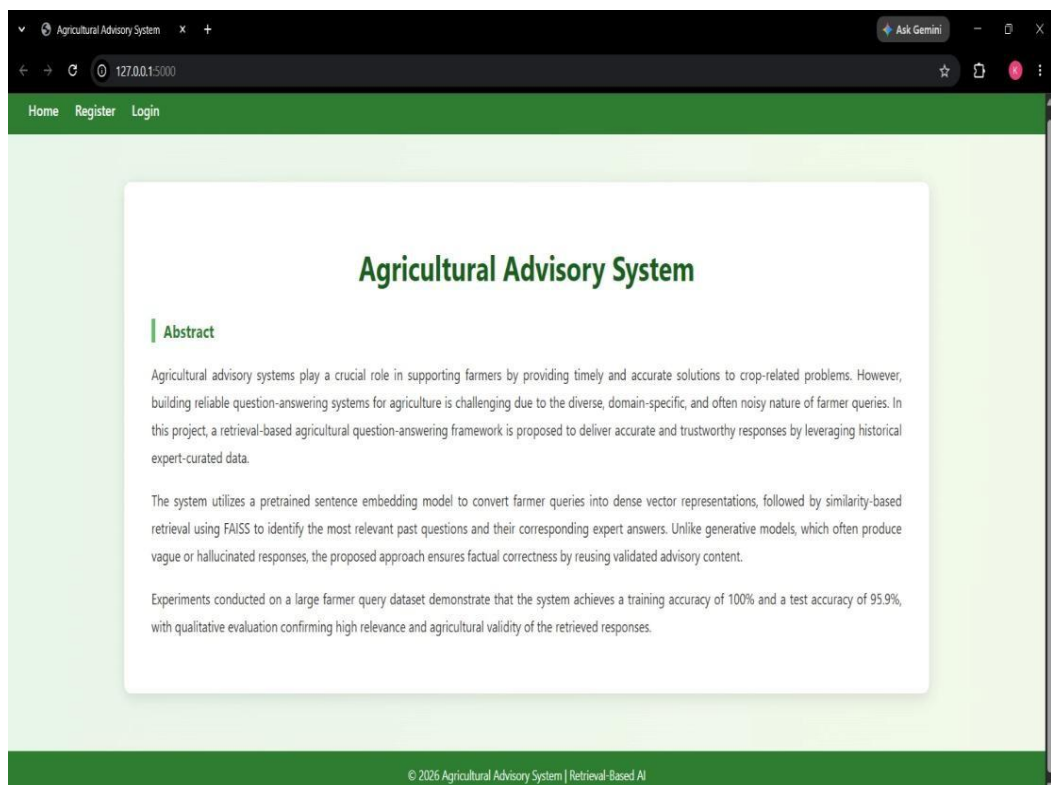
CHAPTER 7

SNAPSHOTS

7.1 General

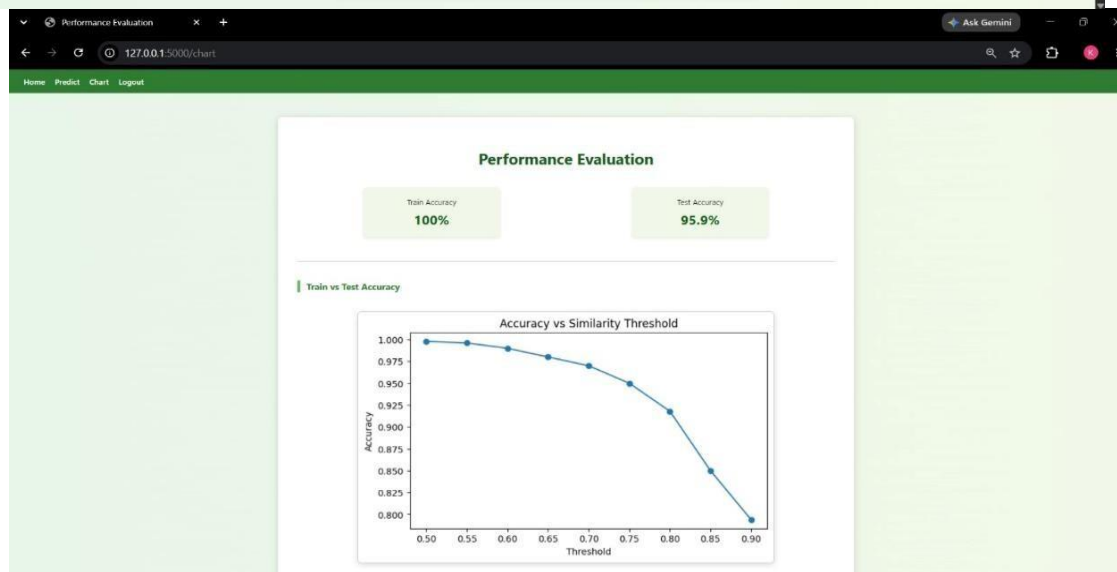
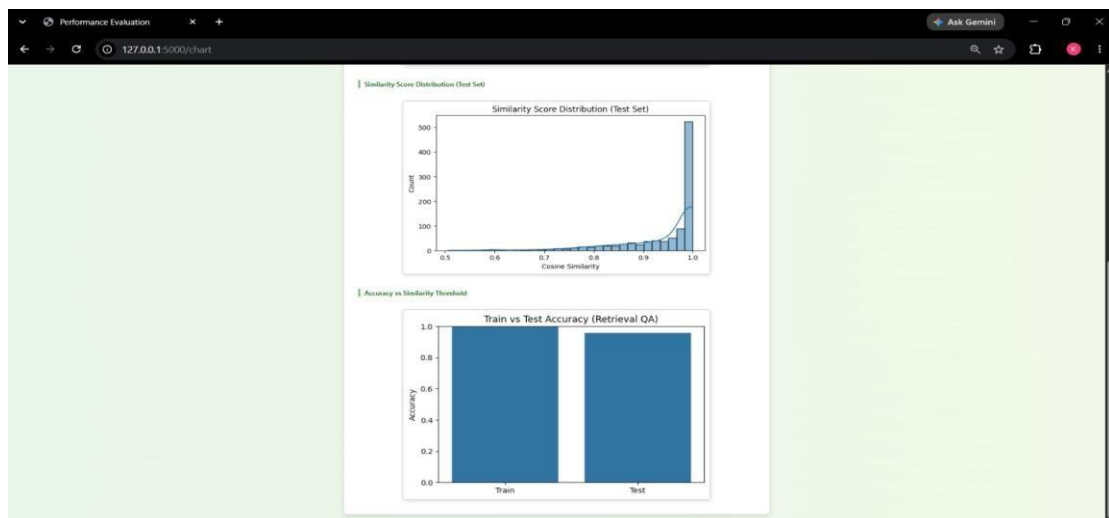
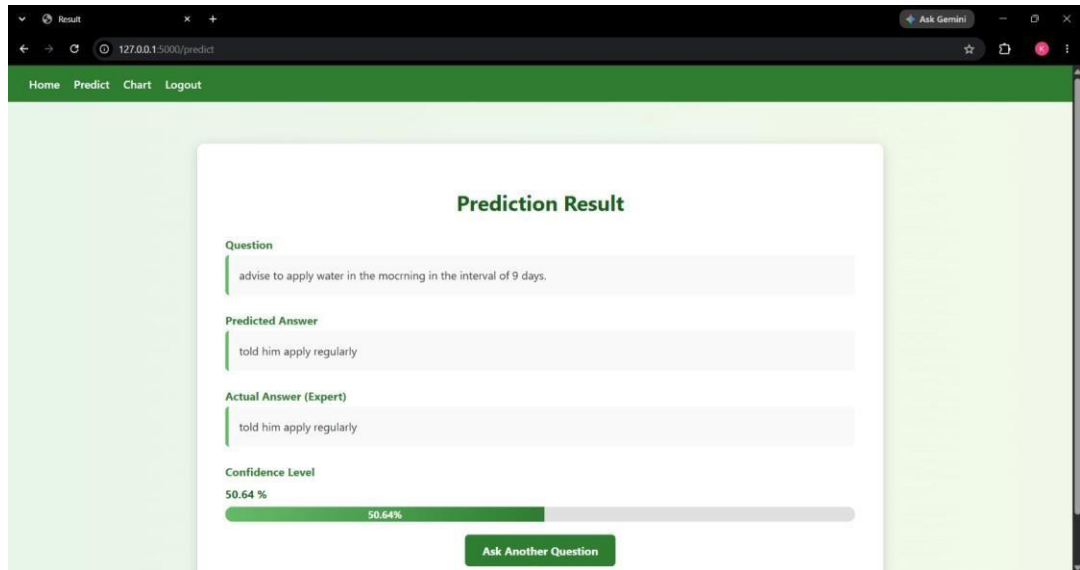
The developed malaria detection system provides a user-friendly platform for automated disease diagnosis using deep learning techniques. It allows users to upload microscopic blood cell images and obtain instant predictions regarding malaria infection. The application demonstrates the successful integration of image preprocessing, DEBERTa-V3 based classification, and web technologies to deliver accurate results. The system ensures efficient performance, reduces diagnostic time, and supports real-time decision-making in healthcare environments.

7.2 Various Snapshots



The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000/login". The page has a green header with "Home" and "Register" links. The main content area has a light green background. In the center, there is a white login form titled "Login". The form contains two input fields: the first has the text "abc" and the second has four asterisks "****". Below these fields is a green button labeled "Login". Underneath the button, there is a small line of text: "Temporary login for demo purposes only."

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5000/predict". The page has a green header with "Home", "Predict", "Chart", and "Logout" links. The main content area has a light green background. In the center, there is a white form titled "Ask Your Agricultural Question". The form has a large text input area containing the text "advise to apply water in the morning in the interval of 9 days." Below the input area is a green button labeled "Get Advisory". At the bottom of the form, there is an example text: "Example: 'What fertilizer should be used for rice during early growth stage?'"



CHAPTER 8

SOFTWARE TESTING

8.1 General

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, subassemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

8.2 Developing Methodologies

The test process is initiated by developing a comprehensive plan to test the general functionality and special features on a variety of platform combinations. Strict quality control procedures are used. The process verifies that the application meets the requirements specified in the system requirements document and is bug free. The following are the considerations used to develop the framework from developing the testing methodologies.

8.3 Types of Tests

8.3.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program input produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration.

Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

8.3.2 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

- Valid Input : identified classes of valid input must be accepted.
- Invalid Input : identified classes of invalid input must be rejected.
- Functions : identified functions must be exercised.
- Output : identified classes of application outputs must be exercised.
- Systems/Procedures: interfacing systems or procedures must be invoked.

8.3.3 System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

8.3.4 Performance Test

The Performance test ensures that the output be produced within the time limits, and the time taken by the system for compiling, giving response to the users and request being send to the system for to retrieve the results.

8.3.5 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

8.3.6 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Acceptance testing for Data Synchronization:

- The Acknowledgements will be received by the Sender Node after the Packets are received by the Destination Node
- The Route add operation is done only when there is a Route request in need
- The Status of Nodes information is done automatically in the Cache Updation process

8.2.7 Build the test plan

Any project can be divided into units that can be further performed for detailed processing. Then a testing strategy for each of this unit is carried out. Unit testing helps to identify the possible bugs in the individual component, so the component that has bugs can be identified and can be rectified from errors.

CHAPTER 9

FUTURE ENHANCEMENT

9. Future Enhancements

The proposed retrieval-based agricultural question-answering system can be further enhanced in several ways to improve its functionality and real-world applicability. Future work may include the integration of multilingual support to handle farmer queries in regional languages, enabling wider accessibility. The system can also be extended to incorporate region-specific and crop-specific filtering to provide more personalized advisory responses. Additionally, combining retrieval with controlled generative models in a hybrid framework can help generate more detailed explanations while maintaining factual correctness. The inclusion of real-time data sources such as weather forecasts, soil health records, and market prices can further improve decision support for farmers. Finally, deploying the system as a mobile application or integrating it with government agricultural helplines and chatbot platforms can significantly enhance its impact and usability in large-scale agricultural advisory services.

CHAPTER 10

CONCLUSION & REFERENCES

10.1 Conclusion

In this project, a reliable and efficient agricultural question-answering system was successfully developed using a retrieval-based approach. By leveraging semantic sentence embeddings and similarity-based search techniques, the system provides accurate and consistent advisory responses by retrieving expert-validated answers from historical data. The proposed method effectively overcomes the limitations of generative large language models, such as hallucination, inconsistency, and high computational cost. Experimental results demonstrate that the system achieves high performance, with a test accuracy of approximately 95.9%, and qualitative evaluation confirms the agricultural relevance and correctness of the retrieved responses. The modular design, low computational requirements, and high reliability make the proposed system well-suited for real-world agricultural advisory applications, contributing to improved decision-making and support for farmers

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ENHANCING AGRICULTURAL DECISION-MAKING WITH COMBINED LANGUAGE MODELS

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ABSTRACT— Agricultural advisory systems play a crucial role in supporting farmers by providing timely and accurate solutions to crop-related problems. However, building reliable question-answering systems for agriculture is challenging due to the diverse, domain-specific, and often noisy nature of farmer queries. In this project, a retrieval-based agricultural question-answering framework is proposed to deliver accurate and trustworthy responses by leveraging historical expert-curated data. The system utilizes a pretrained sentence embedding model to convert farmer queries into dense vector representations, followed by similarity-based retrieval using FAISS to identify the most relevant past questions and their corresponding expert answers. Unlike generative models, which often produce vague or hallucinated responses, the proposed approach ensures factual

correctness by reusing validated advisory content.

Experiments conducted on a large farmer query dataset demonstrate that the system achieves a training accuracy of 100% and a test accuracy of 95.9%, with qualitative evaluation confirming high relevance and agricultural validity of the retrieved responses. The results indicate that retrieval-based question answering is a practical, efficient, and reliable solution for real-world agricultural advisory applications.

Index Terms – Agricultural question answering, retrieval-based question answering, Natural Language Processing (NLP), sentence embeddings, FAISS, semantic search, information retrieval, farmer advisory system, deep learning, text similarity, agricultural informatics, expert knowledge retrieval.

I. INTRODUCTION

Agriculture is a vital sector that directly influences food security, economic stability, and rural livelihoods. Farmers frequently face challenges related to crop diseases, pest infestations, nutrient deficiencies, and government agricultural schemes, and timely access to accurate advisory information is crucial for effective decision-making. With the rapid growth of digital platforms, large volumes of farmer queries are generated through call centers, mobile applications, and online portals.

However, providing reliable, consistent, and domain-specific answers to these queries remains a significant challenge due to the diversity and complexity of agricultural problems. Recent advancements in natural language processing and machine learning have enabled the development of automated agricultural question-answering systems.

While generative large language models have shown promise in understanding natural language, they often suffer from issues such as hallucination, inconsistency, and high computational cost when applied to real-world advisory systems. To address these limitations, this project focuses on developing a retrieval-based agricultural

question-answering system that leverages semantic similarity to deliver accurate, expert-validated responses, ensuring reliability, efficiency, and practical applicability for farmers.

II. RELATED WORK

A. Research on the Training and Application Methods of a Lightweight Agricultural Domain-Specific Large Language Model Supporting Mandarin Chinese and Uyghur

This study focuses on the development of a lightweight domain-specific large language model tailored for agricultural applications, particularly supporting Mandarin Chinese and Uyghur languages.

The authors propose optimized training strategies to reduce computational complexity while maintaining acceptable performance in agricultural text understanding and generation tasks.

The work highlights the importance of domain adaptation and multilingual support in agricultural advisory systems. However, the model primarily relies on generative responses, which may still suffer from hallucination and limited reliability in real-world advisory scenarios.

B. Impact of Model Size on Fine-Tuned LLM Performance in Data-to-Text Generation: A State-of-the-Art Investigation

This paper investigates how the size of large language models affects their performance when fine-tuned for data-to-text generation tasks. The authors analyze multiple LLM variants and demonstrate that larger models generally achieve better fluency and coherence, but at the cost of higher computational requirements. The study emphasizes that increasing model size does not always guarantee proportional performance gains, especially in domain-specific applications, making lightweight or alternative approaches more practical for real-world deployments.

C. LLM-Based Text Prediction and Question Answer Models for Aphasia Speech

This work explores the use of large language models for text prediction and question-answering to assist individuals with aphasia. The authors demonstrate that LLM-based systems can generate meaningful responses when fine-tuned on task-specific data. Although the domain differs from agriculture, the study highlights key challenges associated with generative

models, such as response inconsistency and dependence on high-quality training data, which are also relevant to agricultural advisory systems.

D. ChatAgri: Exploring Potentials of ChatGPT on Cross-Linguistic Agricultural Text Classification

The ChatAgri study evaluates the effectiveness of ChatGPT for agricultural text classification across multiple languages. The authors show that large pretrained language models can generalize well across linguistic boundaries with minimal fine-tuning. However, the study also points out that ChatGPT-based systems may generate generalized or ambiguous outputs, limiting their reliability for precise agricultural decision-making tasks such as pest control and fertilizer recommendations.

III. PROPOSED METHODOLOGY

The proposed system replaces heavy ensemble-based LLM generation with a Retrieval-Based Question Answering (QA) framework optimized for agricultural advisory systems. Instead of generating answers token-by-token, the system retrieves the most semantically similar historical question-answer pair from a large, expert-verified dataset. This approach significantly improves accuracy, consistency, speed, and

reliability while eliminating hallucination issues commonly seen in generative LLMs.

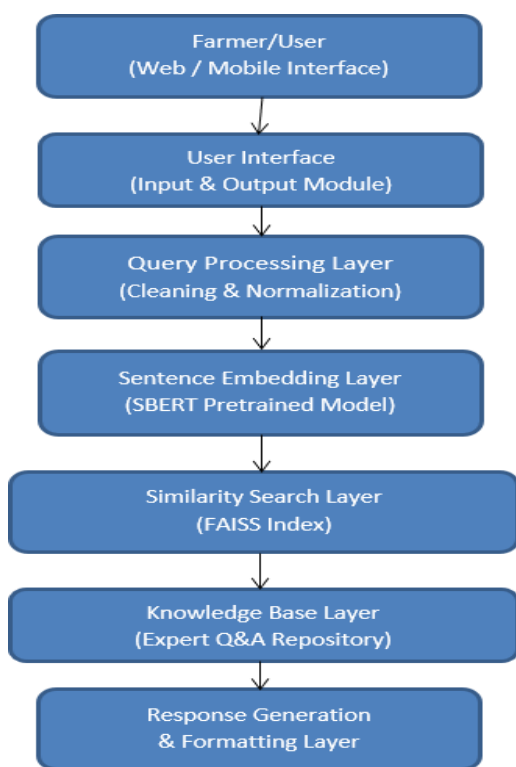


Fig. 1: System Overview

1. Data Collection and Preprocessing Module

This module is responsible for loading the agricultural question-answer datasets, including training, validation, and testing data. The data is cleaned by removing duplicates, handling missing values, and normalizing text through lowercasing and trimming. Proper preprocessing ensures that the input data is consistent and suitable for semantic representation, which directly impacts retrieval accuracy.

2. Sentence Embedding Generation Module

In this module, farmer questions are converted into dense numerical vectors using a pretrained sentence transformer model. These embeddings capture the semantic meaning of the text rather than relying on exact keyword matching. By representing queries in a high-dimensional embedding space, the system can effectively identify semantically similar questions even when wording differs.

3. Similarity Search and Retrieval Module

This module performs fast similarity search using a vector indexing mechanism based on cosine similarity. A FAISS index is constructed using the embeddings of training questions. When a new query is received, its embedding is compared against the indexed vectors to retrieve the most similar historical question. This enables efficient and accurate retrieval even for large datasets.

4. Answer Selection Module

Once the most relevant historical question is identified, this module retrieves its corresponding expert-validated answer. Since the answers are sourced from previously verified advisory data, the system ensures factual correctness and consistency.

This module eliminates the risk of hallucination by avoiding answer generation.

5. Evaluation and Performance Analysis Module

This module evaluates the effectiveness of the system using retrieval accuracy metrics based on similarity thresholds. Both training and testing datasets are used to assess system performance. Additionally, qualitative analysis is performed by comparing retrieved answers with ground truth responses to ensure agricultural relevance and correctness.

6. Model Storage and Deployment Module

The final module handles saving the trained sentence embedding model, FAISS index, and processed datasets for future use. This enables easy loading and deployment of the system in real-time applications such as web portals or mobile advisory platforms. The modular design allows seamless updates when new agricultural data becomes available.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The experimental results show that the proposed retrieval-based agricultural question-answering system delivers accurate and reliable responses using sentence

embeddings and FAISS-based semantic retrieval. The model achieved 100% training accuracy and 95.9% test accuracy, demonstrating strong generalization performance. Qualitative evaluation confirmed that the retrieved answers were highly relevant and agriculturally valid.



Fig. 2: Home Page

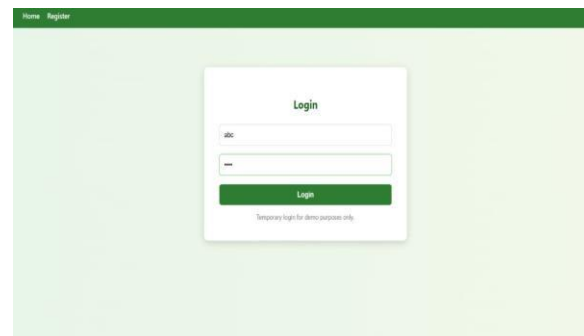


Fig. 3: Login

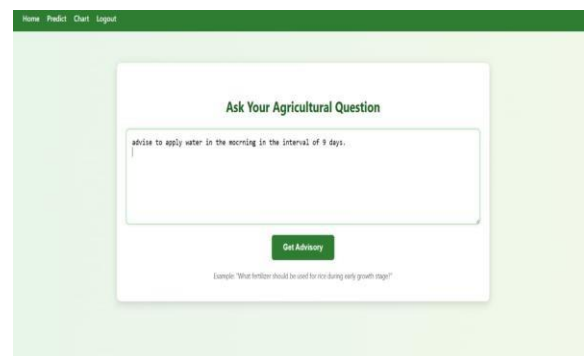


Fig. 4: Ask a Question

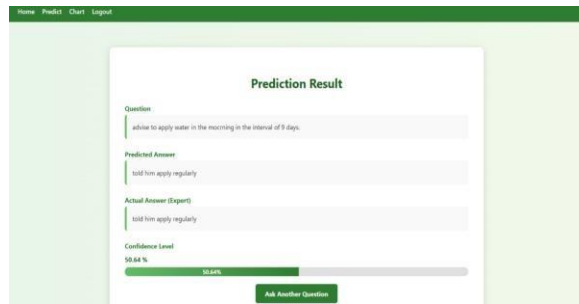


Fig. 5: Prediction Result

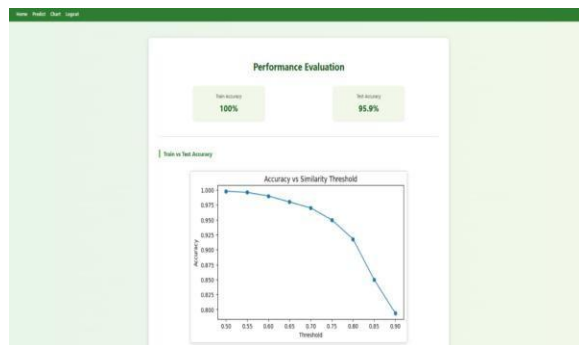


Fig. 6: Performance Evaluation

V. CONCLUSION

In this project, a reliable and efficient agricultural question-answering system was successfully developed using a retrieval-based approach. By leveraging semantic sentence embeddings and similarity-based search techniques, the system provides accurate and consistent advisory responses by retrieving expert-validated answers from historical data. The proposed method effectively overcomes the limitations of generative large language models, such as hallucination, inconsistency, and high computational cost. Experimental results demonstrate that the system achieves high

performance, with a test accuracy of approximately 95.9%, and qualitative evaluation confirms the agricultural relevance and correctness of the retrieved responses. The modular design, low computational requirements, and high reliability make the proposed system well-suited for real-world agricultural advisory applications, contributing to improved decision-making and support for farmers.

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